Mandana Samiei

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Research Focus

My research focuses on understanding the computational principles of intelligent systems, from abstraction to adaptation. I draw on both theoretical analysis and empirical evaluation to build learning algorithms that are capable of efficient learning in complex environments.

Research Highlights — RL, Continual Learning & Agentic Systems

Continual Learning Objectives from a Memory Perspective. Co-led a generalized framework unifying retention- and adaptation-based continual learning methods, under review at TMLR 2026.

Experience Selection for Memory and Reasoning. Co-led the method and experiments on LLM agents testing causal hypotheses, isolating where language agents reproduce human causal-reasoning biases and evaluating interventions to correct them (Cog Sci 2026, COLM 2025; IMOL @NeurIPS 2024).

Abstract Memory Structures for Improved Generalization in RL: Led design and execution of experiments identifying relational memory structures (schemas) that support transfer in RL (Neuron 2026, RLDM 2025, under submission to Nature Communications 2026).

Education

Ph.D. – McGill University & Mila - Québec AI Institute — Montreal, Canada
Advised by Prof. Blake A. Richards & Prof. Doina Precup — *2020 – Present*
Anticipated graduation date: Dec 2026 or earlier

M.Sc. – Concordia University — Montreal, Canada
Advised by Prof. Thomas Fevens — *2017 – 2019*
Thesis: Meta-Learning for Cancer Phenotype Prediction from Gene Expression Data [pdf]

B.Sc. – Shahid Beheshti University — Tehran, Iran
Advised by Prof. Mona Ghassemian — *2012 – 2016*

Selected Research Publications

Presenters are shown in underline. Equal contribution is shown in *. Equal supervision is shown in †.

1. **To Retain or to Adapt? Generalizing Continual Learning.**
   Giulia Lanzillotta*, **Mandana Samiei***, Doina Precup, Razvan Pascanu†, Claire Vernade†.
   Manuscript under review at TMLR, 2026.
2. **Human Adults and LLMs as Scientists: Who Benefits from Active Exploration?**
   Mandana Samiei*, Eunice Yiu*, Anthony GX Chen, Dongyan Lin, Jocelyn Shen, Blake A. Richards, Alison Gopnik†, Doina Precup†. Accepted at Cog Sci 2026. [pdf] [poster]
3. **Language Agents Mirror Human Causal Reasoning Biases. How Can We Help Them Think Like Scientists?**
   Anthony GX Chen , Dongyan Lin*, **Mandana Samiei***, Doina Precup, Blake A. Richards, Rob Fergus, Kenneth Marino. Second conference on Language Modeling (COLM), 2025. [pdf]
4. **Learning Schemas in Reinforcement Learning: bottleneck structure discovery.**
   Mandana Samiei, Doina Precup†, Blake A. Richards†.
   Under submission at Nature Communications, 2026.
5. **The Schema Spectrum: Emergent Structures and levels of abstraction in AI and the Brain.**
   Mandana Samiei, Doina Precup, Blake A. Richards. Accepted at Neuron, 2026. [pdf]
6. **The Role of Schemas in Reinforcement Learning: Implications for Generalization.**
   Mandana Samiei, Doina Precup, Blake A. Richards. Conference on Reinforcement Learning and Decision Making (RLDM), 2025. [pdf]
7. **Local Inconsistency Resolution: The Interplay between Attention and Control in Probabilistic Models.**
   Oliver Richardson, **Mandana Samiei**, Mehran Shakerinava, Joseph Viviano, Abdessamad Kabid, Ali Parviz, Yoshua Bengio. Accepted as a *spotlight* at the twenty-ninth annual conference on artificial intelligence and statistics (AISTATS) 2026. [pdf]
8. **AIF-GEN: Open-Source Platform and Synthetic Dataset Suite for Reinforcement Learning on Large Language Models.** Jacob Chmura*, Shahrad Mohammadzadeh*, Ivan Anokhin, Jacob-Junqi Tian, **Mandana Samiei**, Taz Scott-Talib, Irina Rish, Doina Precup, Reihaneh Rabbany, Nishanth Anand. Championing Open-Source Development in ML Workshop @ ICML, 2025. [OpenReview]

9. **Testing Causal Hypotheses through Hierarchical Reinforcement Learning.**
   Anthony GX Chen*, Dongyan Lin*, **Mandana Samiei***.
   Intrinsically-Motivated and Open-Ended Learning (IMOL) Workshop @ NeurIPS, 2024. [OpenReview]

10. **Mimicking Mammalian Navigation in Watermaze using Brain-Inspired Representations**
    Mandana Samiei*, Arna Ghosh*, Blake A. Richards Biological and Artificial Reinforcement Learning (BARL) Workshop at NeurIPS 2020. [Poster]

11. **Torchmeta: A Meta-Learning Library for PyTorch.**
    Tristan Deleu, Tobias Würfl, **Mandana Samiei**, Joseph Paul Cohen, Yoshua Bengio.
    PyTorch Developer Conference (PTDC), 2019. [arXiv]

Selected Invited Talks and Tutorials

The Role of Schemas in Reinforcement Learning — Oct. 2025
Princeton Reinforcement Learning Lab — *Remote*
- Invited research talk on schema representations in reinforcement learning.

Large Language Models – Tutorial [GitHub] — Sept. 2025
Mediterranean Machine Learning Summer School (M2L) — *Split, Croatia*
- Delivered a tutorial to 100+ participants on the foundations of large language models.

Towards Efficient Generalization in Continual RL using Episodic Memory — Oct. 2021
Microsoft Research Summit 2021 — *Remote*
- Invited talk on memory-augmented RL agents and their generalization efficiency.

RL for Games – Tutorial [Notebook] — Jul. 2021
Neuromatch Academy 2021 — *Remote*
- Created an interactive tutorial on reinforcement learning for games.

Awards and Recognition

Fonds de Recherche du Québec – Nature et Technologies (FRQNT) Doctoral Award	2022-2024
Women in AI Excellence Doctoral Scholarship, Mila	2021
UNIQUE (Unifying Neuroscience & AI) PhD Excellence Scholarship	2020
MITACS Accelerate Research Award	2019
National Merit Scholarship (top 0.2%)	2012

Selected Teaching & Mentorship

Tutor — Large Language Models, Mediterranean ML Summer School (M2L)	2025
Tutor — Introduction to ML, Eastern European Summer School (EEML)	2024
TA — Reinforcement Learning (COMP 579), McGill University	2022
TA — Fundamentals of Machine Learning (IFT 6390), University of Montreal	2021
TA — Intro to Robotics & Intelligent Systems (COMP 417), McGill University	2020

Service & Leadership

Vice President — Women in Machine Learning (WiML)	2026-
EDI Chair and Local Chair — Conference on Lifelong Learning Agents (CoLLAs)	2025, 2024
Board Member — Women in Machine Learning (WiML)	2022–2025
Reviewer — ICLR, NeurIPS, TMLR, CoLLAs, RLC, COLM	2022–2025
Organizer — ML Reproducibility Challenge (MLRC 2023)	2023
Organizer — Mila Neuro-AI Reading Group	2020-2023

Skills

Programming Languages, Libraries, & APIs: Python, PyTorch, Jax, Java, Javascript/CSS/HTML, Bash, Numpy, SciPy, Pandas, Jupyter, Git, SQL, Scikit-learn, TensorFlow, Slurm
Foundational Models, Fine-tuning and Inference: HuggingFace, vLLM, verl, Olama, OpenAI API

References

Prof. Doina Precup – McGill University, Mila, Google DeepMind	dprecup@cs.mcgill.ca
Prof. Blake A. Richards – McGill University, Mila, Google	blake.richards@mila.quebec
Prof. Thomas Fevens – Concordia University	fevens@cs.concordia.ca
Prof. Mona Ghassemian - King's College London, Huawei	mona.ghassemian@kcl.ac.uk